

Predict. Explain. Retain.

A flexible machine-learning platform for customer churn prediction across compatible business datasets.

PREMIUM PROJECT REPORT



Customer churn is expensive because the signal arrives before the cancellation.

ChurnSense AI converts customer records into early warnings and retention priorities.

WHY CHURN MATTERS

- Revenue is lost when high-risk customers leave unnoticed.
- Manual review cannot scale to hundreds or thousands of records.
- Different datasets use different column names, categories and formats.
- A useful model must explain probability, not just return a label.

PLATFORM VALUE

1 - 1000+ AUTO

RECORDS

Single and batch
scoring

CLEANING

IDs and missing
values

LIVE

INSIGHTS

Charts and export

The goal is not just prediction. It is a repeatable decision pipeline for retention teams.

One application supports individual prediction and full dataset analysis.

The interface combines data preparation, model inference and business visualisation.

1

Single Customer

Form-based probability and outcome.

2

Batch Upload

CSV or Excel scoring for large customer sets.

3

Auto Preprocessing

ID removal, missing-value handling and encoding.

4

Feature Alignment

Reorders and fills columns exactly as trained.

5

Analytics

Risk distribution, key metrics and trends.

6

Export

Download scored data for action teams.

Designed for compatible churn datasets, not one hard-coded telecom file.

The system profiles each upload before it allows prediction.

1 Detect target

Recognise likely churn label names.

2 Inspect types

Separate numeric, categorical and identifier columns.

3 Clean data

Handle blanks, nulls and malformed values.

4 Validate schema

Compare required model features with the upload.

COMPATIBILITY CONTRACT

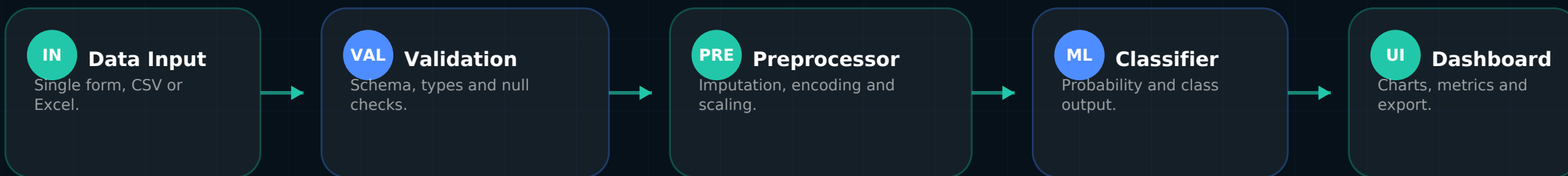
A dataset is compatible when it contains the customer attributes expected by the trained model, or can be mapped into those features.

Flexible rows: one customer or thousands

- Flexible file types: CSV and Excel
- Controlled columns: missing features are handled safely
- Explicit errors: incompatible data is rejected with guidance
-

A complete ML application, not just a notebook model.

The saved preprocessing pipeline is treated as part of the model.



SAVED ARTIFACTS

model_pipeline.joblib
Trained preprocessing + classifier

feature_schema.json
Expected columns and metadata

metrics.json
Validation results and thresholds

Consistent preprocessing is the foundation of reliable churn predictions.

Training and prediction must use the exact same learned transformations.



Why batch prediction once marked every customer as churn.

The issue came from inconsistent categorical encoding and feature alignment between training and upload time.

UNSAFE APPROACH



INCONSISTENT ENCODING

```
01 df[col] = df[col].astype("category").cat.codes
02 # Codes depend on categories present in THIS upload
03 # "Monthly" may be 0 in one file and 2 in another
```

The model receives different numbers for the same category, so batch outcomes become distorted.

CORRECT APPROACH



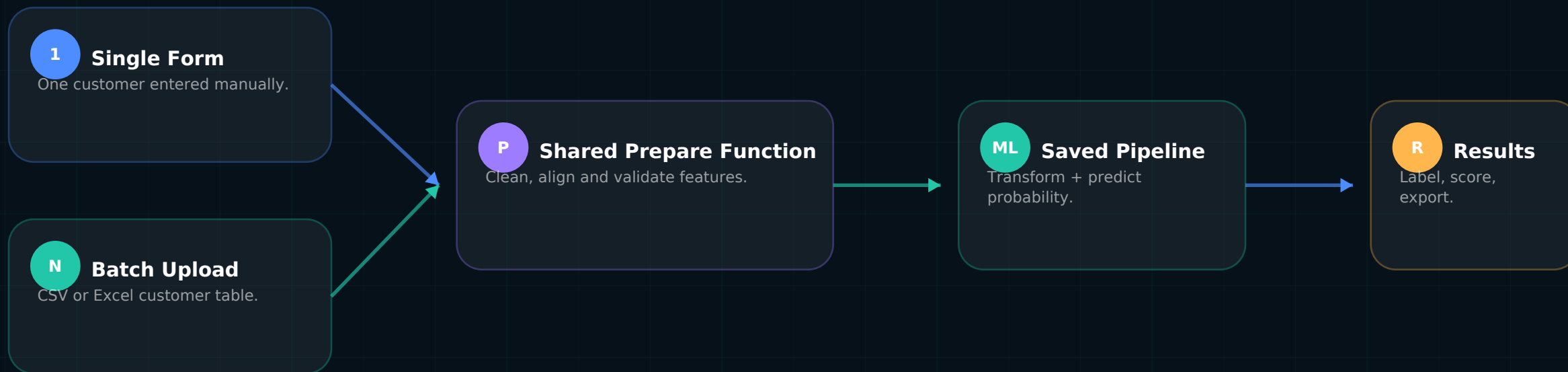
CONSISTENT PIPELINE

```
01 pipeline = joblib.load("model_pipeline.joblib")
02 X = align_to_training_schema(uploaded_df)
03 prob = pipeline.predict_proba(X)[:, 1]
04 pred = prob >= decision_threshold
```

The fitted encoder and model travel together, guaranteeing the same transformation for single and batch data.

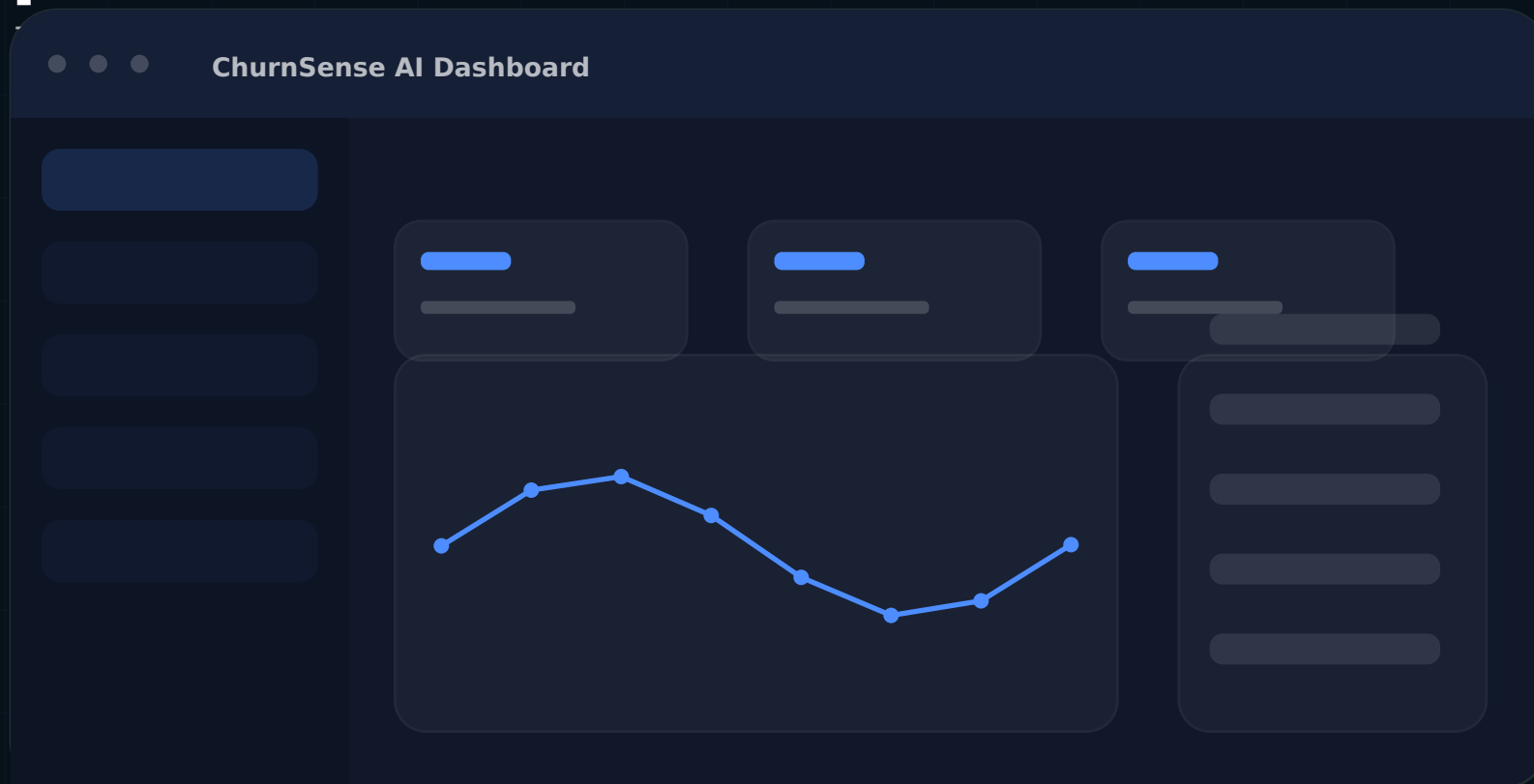
Single-customer and batch modes now share one inference path.

Only the input interface changes; validation, preprocessing and model logic remain identical.



This architecture prevents separate code paths from silently drifting apart and ensures that a customer receives the same prediction in both modes.

Predictions become useful when teams can prioritise action.



KEY OUTPUTS

Total customers processed

- Predicted churners and retention pool
- Average probability and high-risk segment
- Downloadable scored file for outreach
- Distribution charts for rapid presentation

A churn model must be tested on mixed outcomes, not a single obvious case.

Testing covers schema behaviour, probability quality and consistency across interfaces.

OK

Mixed test set

Contains both churn and no-churn records.

OK

Interface parity

Same row gives same result in single and batch modes.

OK

Column order

Predictions remain stable after upload reordering.

OK

Missing values

Imputation handles blanks without crashes.

OK

Unknown categories

Encoder ignores or maps unseen values safely.

OK

Threshold review

Business threshold is documented and adjustable.

The next level is repeatable retraining, versioning and monitoring.

ChurnSense already has the right foundation: a saved end-to-end pipeline and validated upload flow.

Schema

DATA CONTRACT

Reject incompatible uploads with clear messages.

Versioned

MODEL FILES

Track pipeline, features, metrics and date.

Logged

PREDICTIONS

Store probability and model version for audits.

Monitored

DRIFT

Watch changes in inputs and outcome quality.

SECURE DELIVERY CHECKLIST

Restrict upload size and supported file formats.

- Never log sensitive customer fields unnecessarily.
- Pin package versions and validate loaded model artefacts.
- Add authentication before storing company datasets.
-

From a college ML project to a practical retention intelligence product.

Future improvements should focus on explainability, retraining and business action loops.

NOW

Robust single and batch prediction with automatic preprocessing.

NEXT

SHAP explanations, configurable target mapping and model comparison.

LATER

Automated retraining, CRM integration and retention campaign tracking.

Project statement: ChurnSense AI demonstrates end-to-end machine learning engineering: data profiling, reusable preprocessing, model persistence, reliable batch inference and decision-ready visualisation.